

The background of the slide features a stylized brain shape in the upper half, filled with intricate circuit board patterns and glowing nodes. Below the brain, a cityscape is visible, with a prominent building in the center that appears to be emitting a signal or light, connecting the brain to the city. The overall color palette is dark and muted, with greys, blues, and browns, accented by the glowing elements of the circuitry.

MediSyn/BRIDGE

Group 2:

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IDEA



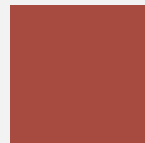
Interprofessional Education (IPE) is a collaborative approach that brings together students from various disciplines to learn collaboratively, aiming to enhance teamwork.



Interprofessional Practice (IPP) is putting it into practice.



BRIDGE delves into building a Multi-Agent chatbot system capable of simulating educational conversations representing diverse professions in the medical and public health setting.

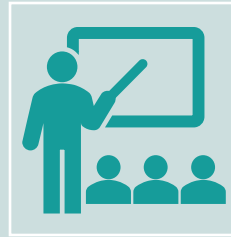


MediSyn is a multi-agent system that processes a patient's IPE case study, retrieves the appropriate healthcare roles, and generates individualized treatment plans from each profession's perspective.

Current Issues



SCHEDULING: DIFFERENT PEOPLE HAVE DIFFERENT LOGISTICS, MAKING IT HARD TO FIND TIME FOR JOINT SESSIONS.



FINANCIAL CONSTRAINT: COORDINATING IPE REQUIRES EXTRA FUNDS FOR THINGS LIKE FACULTY TIME, SIMULATION TOOLS, AND SHARED SPACES.



CULTURAL BARRIERS: STEREOTYPES AND HIERARCHICAL POWER IMBALANCES BETWEEN PROFESSIONS CAN BLOCK EFFECTIVE TEAMWORK.

OVERVIEW



MediSyn

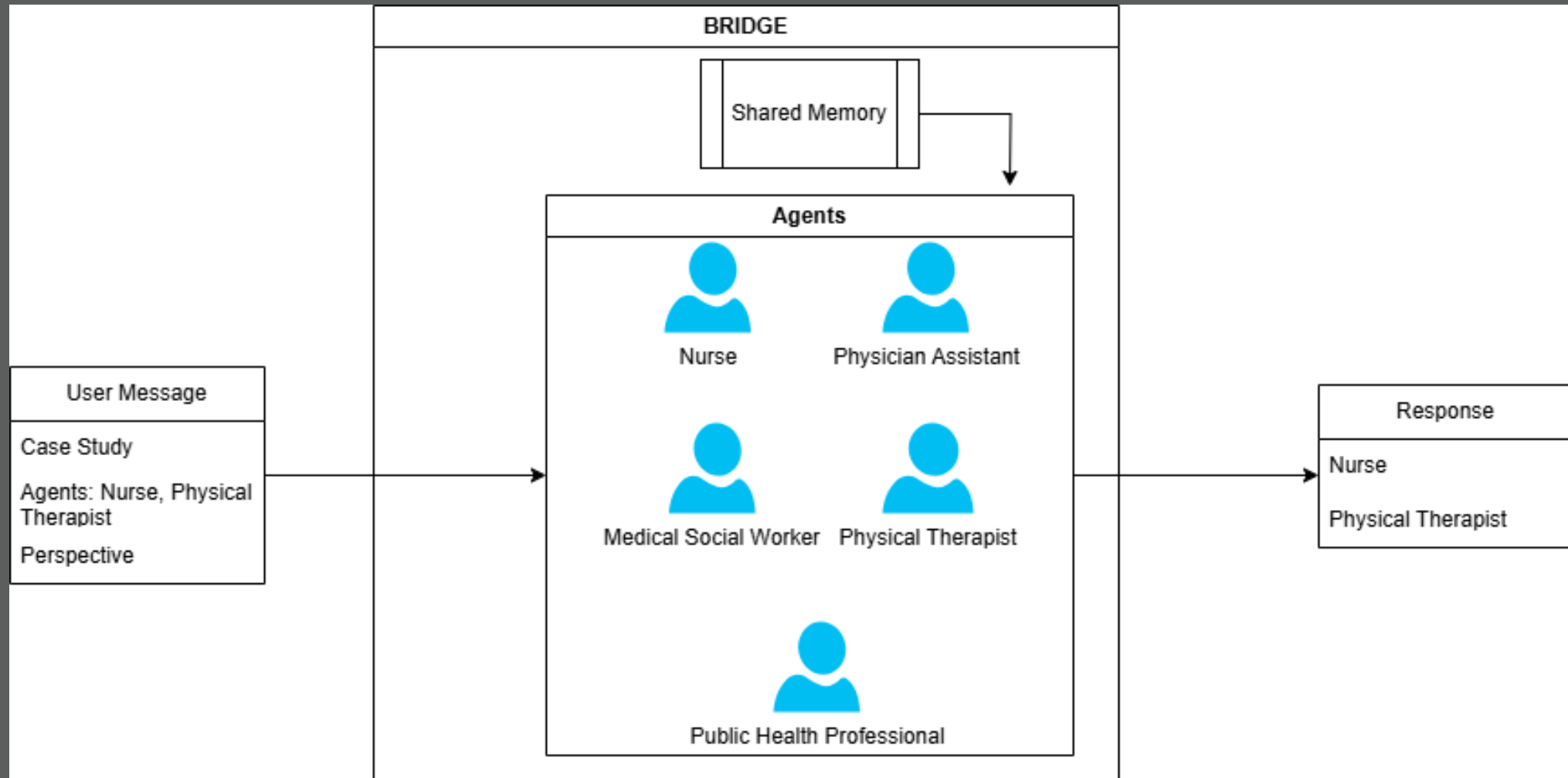
Data about healthcare IPP patient cases
Build Multi-agent system that extracts 'Team Members' and builds catered IPP treatment plan by the various team members who are healthcare professionals from various disciplines.



BRIDGE

Specific private case studies
Build Multi-agent system, that has specific agents from various disciplines, to facilitate IPE.
Build a website for customer use.

BRIDGE- Architecture



BRIDGE- Professions

Agents:

- Nurse
- Physician Assistant
- Medical Social Worker
- Physical Therapist
- Public Health Professional

The idea is that given a patient's case study, these agents should be able to substitute for professionals and help collaborate and facilitate learning by being able to answer various questions regarding the case.

BRIDGE Webapp

Select Agents



Nurse



Physician Assistant



Medical Social Worker



Physical Therapist



Public Health Professional

Select All Agents

BRIDGE: Multi-Agent IPE Platform

Ask questions to collaborate with various healthcare/public health professionals.

 Clear Chat

Current case:

Opioid Overdose and Psychiatric Admission - Julia Baker

What do you think of Julia?



Nurse

Julia is facing a challenging situation that requires empathetic and comprehensive care. Her past struggles with social anxiety, ADHD, and recent job loss may contribute to her substance use. The current opioid overdose is a critical wake-up call for needed intervention. Our focus will be on stabilizing her physical health first, then ensuring she receives the appropriate psychiatric and addiction support. It's important to involve her in a supportive environment, perhaps engaging her in therapy and exploring community resources. Addressing both her mental health and addiction recovery needs will be critical in her recovery journey. We have to support her through this process.

Who

5/500

Ask 4 Agents



MediSyn

- MediSyn is a multi-agent system that generates a role-specific IPP treatment plan for a patient's case.
- Existing AI chatbots typically generate treatment plans from a single-provider perspective.
- They lack role-specific care that's needed for patient's care in IPE.
- Goals:
 - *Where* a multi-agent system like MediSyn fits in the broader landscape of AI-assisted treatment planning
 - *How* it compares to single-agent (single-role) systems and prompt tuned

MediSyn: Process

Convert all the data from
PDFs to Stuctured JSONs

Pass the Data to LLM &
Multi-Agent System to
generate Agent specific
Treatment Plan
representative of Profession
Specific Treatment Plan

Evaluate performance using
metrics (BLEU, ROUGE
score, etc.)

Compare results generated from
single LLM vs. Multi-age

MediSyn: Data Extraction

Extract text from
structured PDF
using
UNSTRUCTURED

System Prompt
instructs LLM to
extract relevant
fields

LLM is invoked to
parse response and
save to JSON

We also do a manual extraction of the data to have a JSON ground truth dataset

MediSyn: Raw Data

- Data: PDFs with the patient case, IPP team members, assessment plans and results, IPP treatment plans.
- 15 case studies used.

Case Study:

Collaboration Helps Clinical Team Diagnose Genetic Disorder

SIG 5: Craniofacial and Velopharyngeal Disorders

Summary

An interprofessional practice (IPP) team at a cleft lip and palate clinic developed a treatment plan for a 4-year-old girl with hypernasal speech and dysmorphic features. By working across specialties, the team diagnosed the girl with 22q11.2 deletion syndrome (22q11.2DS) and executed a treatment plan that included speech-language therapy and surgery to create a pharyngeal flap.

Patient Info



ELIZABETH
4-YEAR OLD

Current Diagnosis:
22q11.2 deletion syndrome

Meet The Team



Audiologist



Ear, nose, & throat
specialist (ENT)



Geneticist



Nurse
practitioner (NP)



Plastic
surgeon



Psychologist



SLP



Social worker



Patient



Family

MediSyn: Extracted Data

Field	Description
Patient_info	Includes patient's name, age, diagnosis
Summary	Summary of the case
Team_members	List of involved professionals
Background	Patient's background
Assessment_plan	Evaluation responsibilities assigned to each professional
Assessment_results	Results of the diagnostic evaluations
Treatment_plan	Final IPP treatment plan co-developed by the team

MediSyn: Approaches

LLMs:

LLAMA 3.2 70b

Claude 3.5 sonnet

GPT 4o

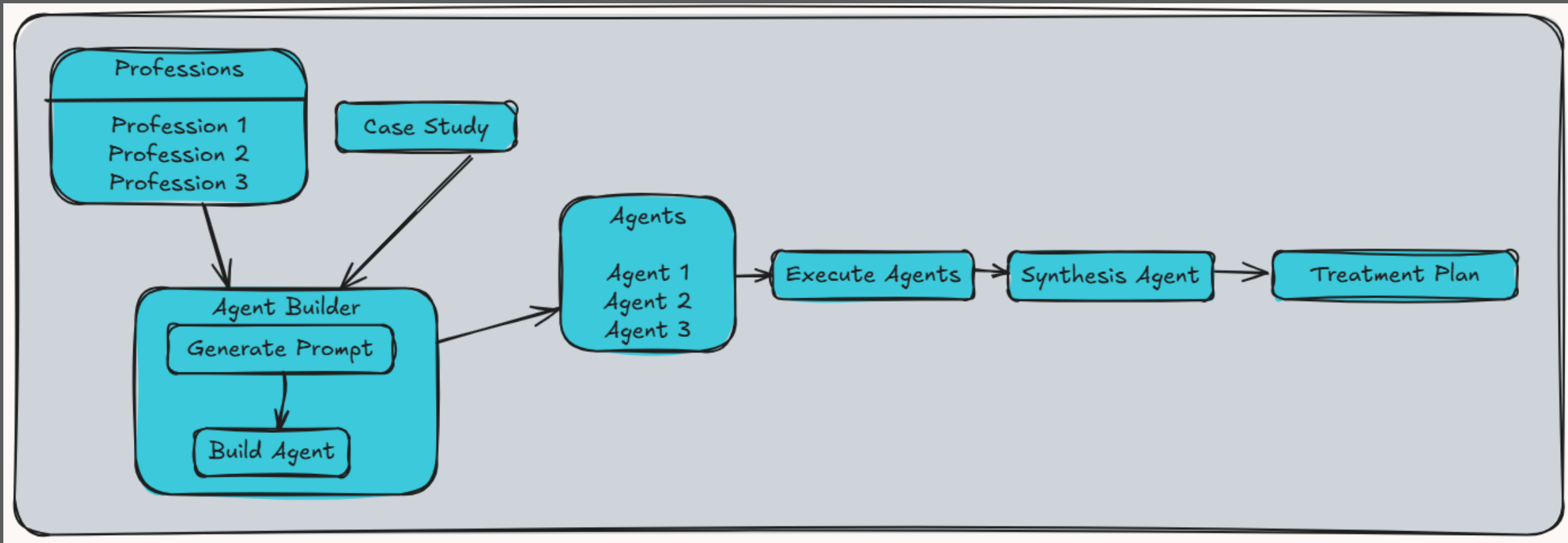
Baseline approach: Uses Single LLM with a simple prompt

Prompt-tuned Approach: Carefully crafted prompt to generate role specific treatment plan

MediSyn: Proposed Multi-agentic approach

We compare the results for both manually prepared data and LLM prepared Data

MediSyn: Architecture



MediSyn: Evaluation



Quantitative:

BLEU Score- Measures n-gram overlap between generated and reference treatment plans

ROUGE score- assesses how much of the reference plan content is captured in the generated plans

LLM-as-a-judge evaluation (GPT- 4o):

- Accuracy
- Plausibility
- Specificity
- Omission



Qualitative:

Manual reading of treatment plans
(structure, role attribution,
collaborative language)

MediSyn: Results (Data Preparation)

```
"patient_info": {  
  "name": "Elizabeth",  
  "age": "4 year old",  
  "diagnosis": [  
    "22q11.2 deletion syndrome"  
  ]  
}
```

Patient Info: Manual

```
"patient_info": {  
  "name": "Elizabeth",  
  "age": "4-year-old",  
  "diagnosis": [  
    "22q11.2 deletion syndrome"  
  ]  
}
```

Patient Info: GPT

```
"patient_info": {  
  "name": "Elizabeth",  
  "age": "4-year-old",  
  "diagnosis": [  
    "22q11.2 deletion syndrome  
(22q11.2DS)",  
    "hypernasal speech",  
    "dysmorphic features",  
    "articulation disorder",  
    "velopharyngeal disorder (VPD)"]  
}
```

Patient Info: Llama

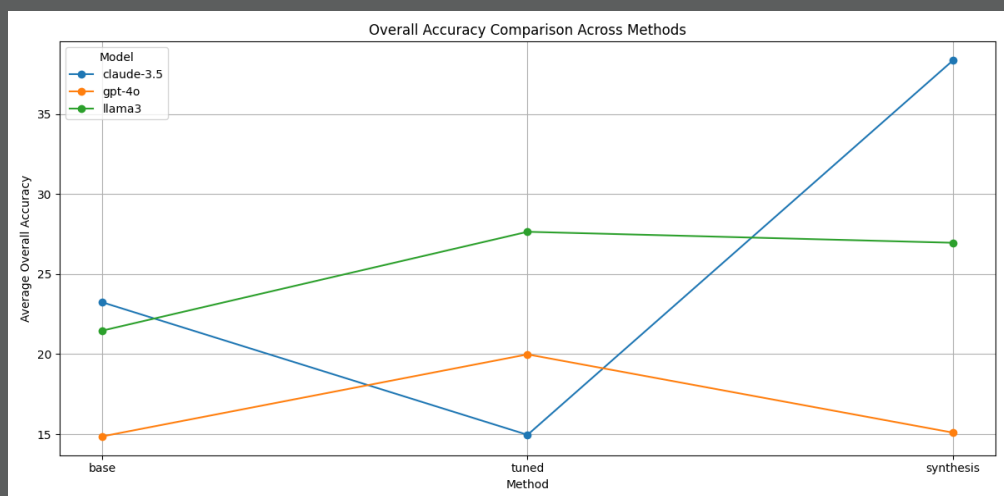
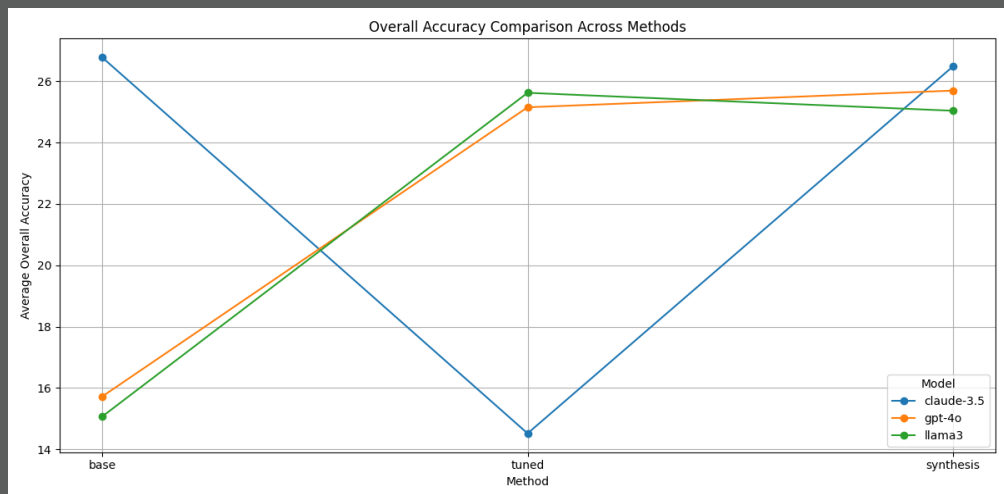
```
"patient_info": {  
  "name": "Elizabeth",  
  "age": "4-year-old",  
  "diagnosis": [  
    "22q11.2 deletion syndrome",  
    "Velopharyngeal disorder",  
    "Severe articulation disorder",  
    "Severe hypernasality"  
  ]  
}
```

Patient Info: Claude

- GPT preserved the format closest to the manual version
- Claude and Llama included additional conditions in diagnosis
- Overall required data is captured

The case is similar in other fields: GPT stays true to exact data. Claude and Llama reformat into more prose-driven structure

MediSyn: Results (Quantitative Analysis)



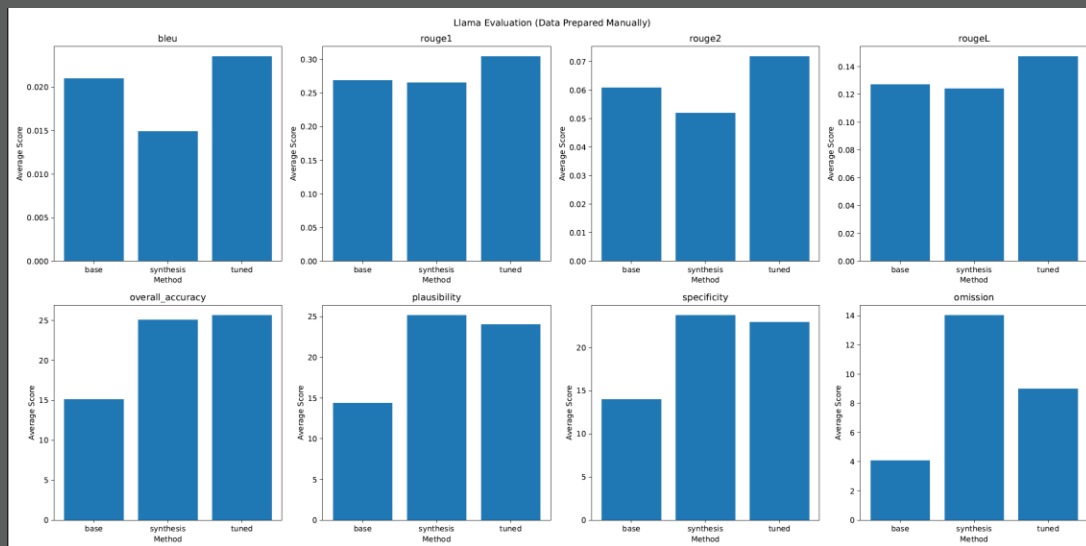
LLM-as-a-judge Overall Accuracy

- Manual data consistently enhances overall accuracy
- Claude performs best in MediSyn
- Surprisingly, Claude prompt-tuned shows a massively low overall accuracy
- Llama has highest accuracy in prompt-tuned version.

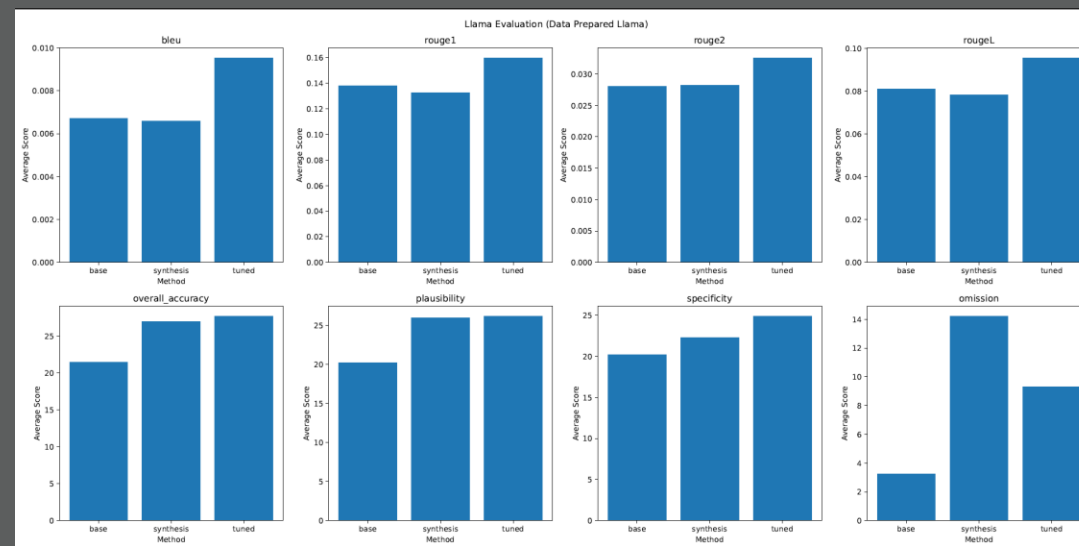
- LLM data shows variable results
- Claude performs best in MediSyn by a huge margin. Claude prompt-tuned drops from baseline here as well.
- GPT and Llama show best accuracy in prompt-tuned version

MediSyn: Results (Quantitative Analysis)

MANUAL



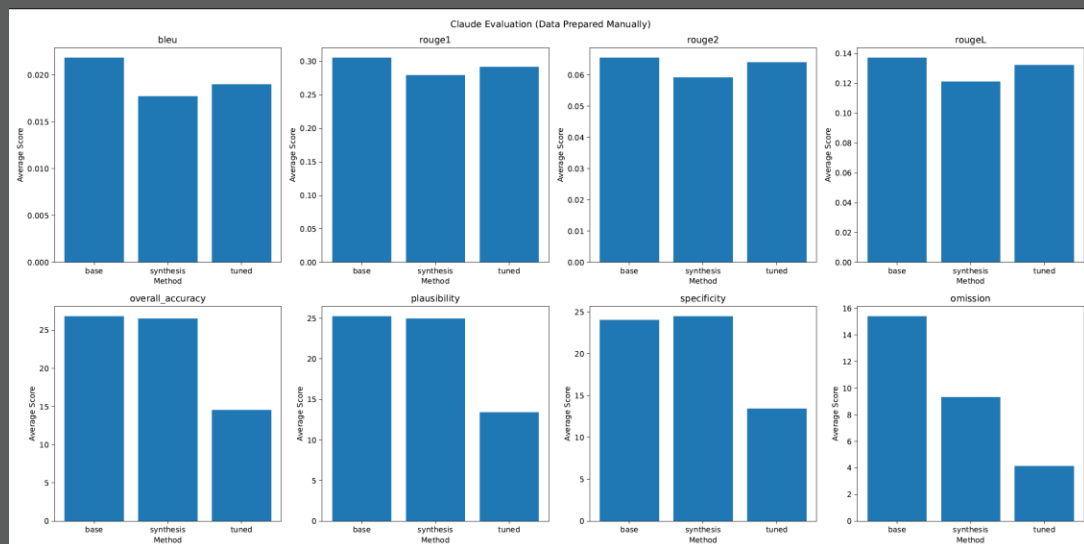
LLM - LLAMA



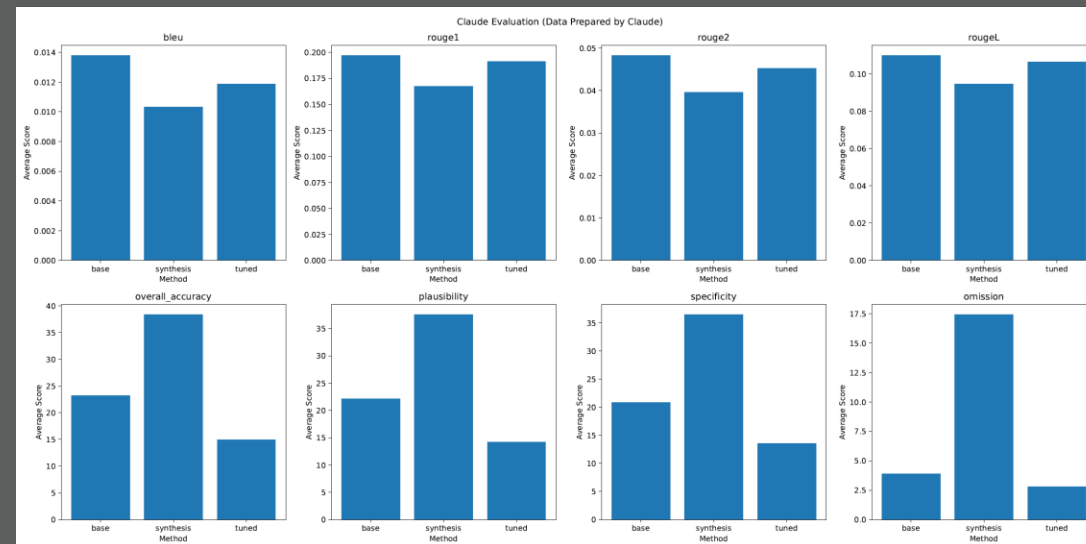
- In terms of BLEU and ROUGE scores, prompt tuned version performs best for both.
- In LLM-as-a-judge metrics, it's hard to choose between prompt-tuned & MediSyn

MediSyn: Results (Quantitative Analysis)

MANUAL



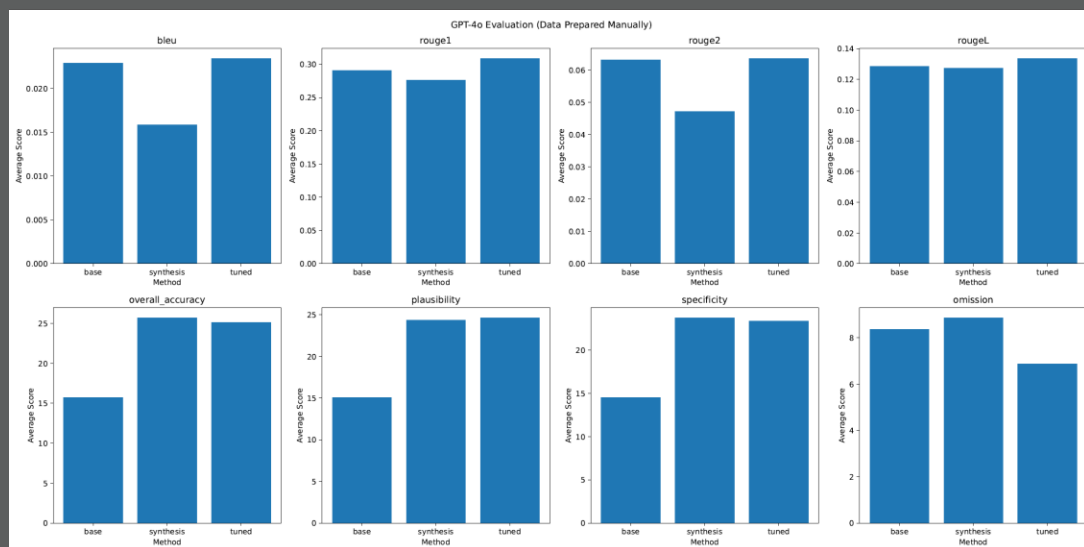
LLM - CLAUDE



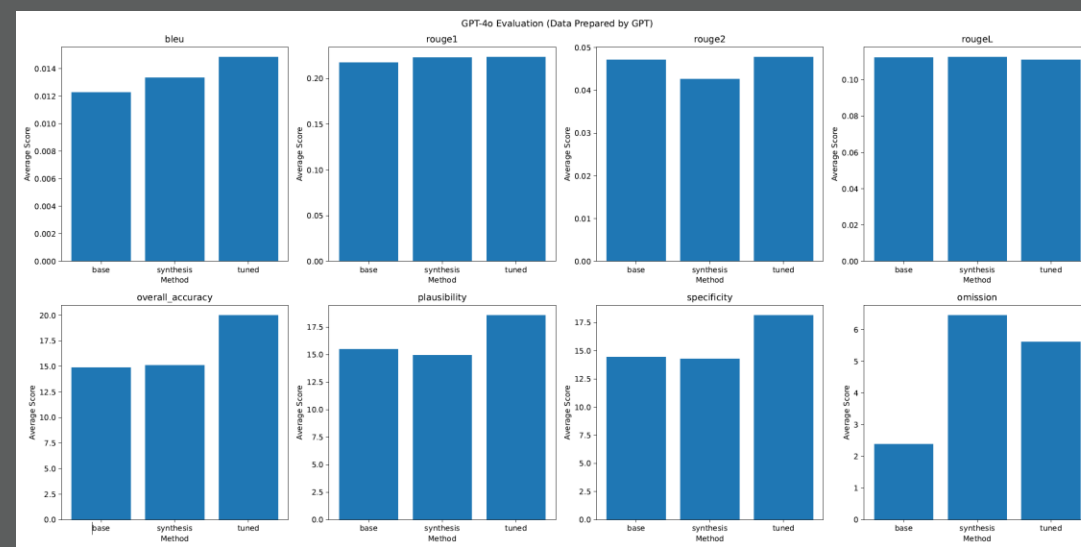
- In terms of BLEU and ROUGE scores, surprisingly, base model performs the best.
- In LLM-as-a-judge metrics, for manual data preparation, it's hard to choose between base version and MediSyn
- For LLM prepared data, MediSyn is the clear winner.

MediSyn: Results (Quantitative Analysis)

MANUAL



LLM - GPT4O



- For BLEU and ROUGE scores, we see a drop in MediSyn's performance.
- In LLM-as-a-judge, MediSyn is the best for manual data, while Prompt-tuning takes it for LLM data.

MediSyn: Results (Qualitative Analysis)

- **MANUAL Llama:**
 - Baseline llama lacks role attribution and shows redundancy
 - Prompt-tuned improves clarity and IPE alignment with role specific structuring (best)
 - Medisyn gives polished but organized into discipline-based sections not individual roles
- **MANUAL Claude:**
 - Baseline claude lacks role attribution; like a clinical summary
 - Prompt-tuning Improves role level clarity and IPE alignment, although verbose
 - Medisyn Claude improves cohesiveness and reduces verbosity (best)
- **MANUAL GPT:**
 - Baseline is concise but lacks apt roles. Has roles like "frequency", "goals", "purpose".
 - Prompt-tuned response is quite good and hits all the checkmarks
 - Medisyn performs even better with better tonality and an addition of "Overall coordination and follow-up" (best)

"generated_text" : "***Comprehensive Treatment Plan for Elizabeth**\n\n*I. Speech-Language Therapy**\n\n* Goal: Improve speech intelligibility and address hypernasal speech\n\nInterventions:\n\n\t+ Weekly speech-language therapy sessions with a certified SLP\n\n\t+ Focus on articulation, resonance, and language skills\n\n\t+ Use of visual aids, mirrors, and oral motor exercises to improve speech production\n\n\t+ Parent-child interaction therapy to enhance communication skills\n\n\n**II. Surgical Intervention**\n\n* Goal: Create a pharyngeal flap to address velopharyngeal dysfunction\n\nInterventions:\n\n\t+ Collaborate with the plastic surgeon to schedule surgery\n\n\t+ Pre-surgical evaluation and preparation with the SLP and surgeon\n\n\t+ Post-surgical speech therapy to address any changes in speech production\n\n\n**III. Genetic Counseling and Medical Management**\n\n* Goal: Confirm diagnosis of 22q11.2 deletion syndrome and address potential medical implications\n\nInterventions:\n\n\t+ Genetic counseling with the geneticist to discuss diagnosis and implications\n\n\t+ Referral to specialists (e.g., cardiologist, immunologist) as needed\n\n\t+ Regular medical check-ups to monitor overall health and address any related medical issues\n\n\n**IV. Audiological Management**\n\n* Goal: Monitor hearing and middle ear function\n\nInterventions:\n\n\t+ Regular audiological assessments to monitor hearing and middle ear function\n\n\t+ Referral to ENT specialist if any concerns arise\n\n\n**V. Psychological and Social Support**\n\n* Goal: Address developmental and behavioral concerns\n\nInterventions:\n\n\t+ Referral to a psychologist for developmental and behavioral assessment\n\n\t+ Parent-child interaction therapy to enhance communication skills and address any behavioral concerns\n\n\t+ Social work support to address any barriers to care and provide community resources\n\n\n**VI. Interdisciplinary Team Collaboration**\n\n* Goal: Ensure comprehensive care and coordination among team members\n\nInterventions:\n\n\t+ Regular team meetings to discuss progress, concerns, and recommendations\n\n\t+ Collaboration among team members to develop and implement a comprehensive treatment plan\n\n\n**Family Education and Support**\n\n* Goal: Educate and support Elizabeth's family\n\nInterventions:\n\n\t+ Education on 22q11.2 deletion syndrome, its implications, and management\n\n\t+ Training on strategies to facilitate communication and social skills\n\n\t+ Ongoing support and counseling to address emotional and psychological concerns\n\n\nBy working together as an interprofessional team, we can provide comprehensive care to Elizabeth, addressing her complex needs and promoting optimal outcomes in speech, language, and overall development."

MediSyn: Results (Qualitative Analysis)

- **LLM Llama:**
 - Baseline llama covered the core roles (SLP, Audiologist) but adds unnecessary sections ("Developmental Monitoring"); lacks nuance
 - Prompt-tuned shows clear roles
 - Medisyn gives structured, role-specific and comprehensive responses (some over-elaboration) (best)
- **LLM Claude:**
 - Baseline claude lacks role segmentation
 - Prompt-tuning introduces roles although slightly redundant
 - Medisyn Claude is domain specific, has actionable strategies and collaborative, tone is empathetic (best)
- **LLM GPT:**
 - Baseline lacks role-specificity; vague recommendations
 - Prompt-tuned response is role-specific but very compressed and non-elaborative
 - Medisyn gives great interprofessional depth, even combines overlapping roles to present singular solution (best)

"generated_text" : "{}\n\n**Audiologist:**\n- Conduct regular audiological evaluations every 6 months to monitor hearing status.\n- Educate family on the significance of optimal hearing for speech development and signs of ear infections.\n\n**Speech-Language Pathologist (SLP):**\n- Provide intensive speech-language therapy twice weekly, focusing on articulation, resonance exercises, and oral-motor skills.\n- Use visual and tactile feedback techniques to improve speech production.\n- Involve parents in exercises to facilitate practice at home.\n\n**ENT Specialist:**\n- Regular ENT evaluations to monitor ear health and address infections.\n- Tympanometry checks every 6 months and management of middle ear concerns.\n- Collaborate on pre- and post-surgical evaluations for pharyngeal flap surgery.\n\n**Surgeon:**\n- Plan and perform pharyngeal flap surgery within 6 months to address velopharyngeal insufficiency.\n- Provide post-surgical follow-up care in collaboration with the SLP to assess outcomes.\n\n**Psychologist:**\n- Conduct regular assessments of developmental and behavioral health.\n- Provide support for social development through play therapy and peer interaction activities.\n\n**Geneticist:**\n- Offer genetic testing and counseling to confirm and understand 22q11.2DS.\n- Provide information on inheritance patterns and family planning implications.\n\n**Nurse Practitioner (NP):**\n- Coordinate medical care, monitor growth, and manage vaccinations.\n- Conduct regular developmental screenings and assessments for emerging needs.\n\n**Social Worker:**\n- Address barriers to care, such as transportation or financial concerns.\n- Connect the family with community resources, support groups, and financial aid programs.\n\n**Interprofessional Team Coordination:**\n- Hold bi-monthly meetings to track progress and adjust the treatment plan.\n- Maintain clear communication across all team members and with Elizabeth's family.\n\n**Family Involvement:**\n- Educate the family about 22q11.2DS and engage them in therapy sessions.\n- Provide educational materials and support for home practice and involvement in Elizabeth's care.\n\nBy following this comprehensive treatment plan, Elizabeth's team aims to improve her speech, support her overall development, and provide necessary resources to her family, ensuring a coordinated, holistic approach to her care."

Conclusion

- Quantitative results are confusing and suggests that maybe just a prompt-tuning is sufficient.
- Qualitative results make it clear that the Agentic system- MediSyn gives the best results.
- *To what extent should we rely on traditional quantitative metrics when evaluating clinical reasoning outputs, and when should human-aligned criteria take precedence?*
- In order to actually verify the treatment plan's efficiency, real human profession-specific analysis would be required.

THANKS